T3. PN junctions in semiconductors — Solution

A1. Let A be the contact area. The volume charge density is $\rho = \pm q_e N_{a,d}$ ($+$ for holes, $-$ for electrons), so the condition that the total charge of the holes and electrons is zero means $AN_a d_p = AN_d d_n$. Therefore

$$\boxed{N_a d_p = N_d d_n.}$$

A2. By Gauss's law in differential form,

$$\operatorname{div}\vec{E} = \frac{\rho}{\varepsilon_0 \varepsilon}.$$

Since $\vec{E} = \vec{E}(x)$,

$$\operatorname{div}\vec{E} = \frac{dE_x}{dx},$$

which proves the given formula.

A3. Using the result of A2 together with the relation between the concentration and the volume charge density, $\rho = \pm q_e N_{a,d}$ ($+$ for holes, $-$ for electrons), we obtain

$$\frac{dE_x}{dx} = \begin{cases} -\dfrac{q_e N_a}{\varepsilon_0 \varepsilon}, & -d_p \le x \le 0, \\ \dfrac{q_e N_d}{\varepsilon_0 \varepsilon}, & 0 < x \le d_n, \\ 0, & x < -d_p \text{ or } x > d_n. \end{cases}$$

Now we need the indefinite integral and the constant of integration. To get the answer faster, let us compute the definite integral directly. Since $E_x(\pm\infty) = 0$,

$$E_x(x) = \int_{-\infty}^{x} E'(x)\,dx = -\int_{x}^{+\infty} E'(x)\,dx.$$

Using these relations, we obtain the final answer:

$$\boxed{E_x(x) = \begin{cases} -\dfrac{q_e N_a (x + d_p)}{\varepsilon_0 \varepsilon}, & -d_p \le x \le 0, \\ \dfrac{q_e N_d (x - d_n)}{\varepsilon_0 \varepsilon}, & 0 < x \le d_n, \\ 0, & x < -d_p \text{ or } x > d_n. \end{cases}}$$

A4. In one dimension $\varphi'(x) = -E_x(x)$. With the condition $\varphi(0) = 0$, the electrostatic potential can be written as

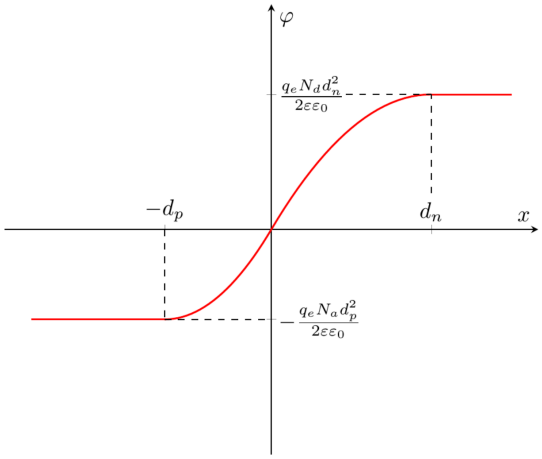
$$\varphi(x) = -\int_0^x E_x(x)\,\mathrm{d}x.$$

Computing the integral directly, we obtain

$$\boxed{\varphi(x) = \begin{cases} -\dfrac{q_e N_a d_p^2}{2\varepsilon_0 \varepsilon}, & x < -d_p, \\ \dfrac{q_e N_a (x^2 + 2d_p x)}{2\varepsilon_0 \varepsilon}, & -d_p \le x < 0, \\ \dfrac{q_e N_d (2d_n x - x^2)}{2\varepsilon_0 \varepsilon}, & 0 \le x < d_n, \\ \dfrac{q_e N_d d_n^2}{2\varepsilon_0 \varepsilon}, & x \ge d_n. \end{cases}}$$

$$\boxed{\Delta\varphi = \frac{q_e \left(N_a d_p^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0}.}$$

A5.



A qualitative graph of the potential $\varphi(x)$.

A6. From A4,

$$\Delta\varphi = \frac{q_e \left(N_a d_p^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0}.$$

Using the relation $N_a d_p = N_d d_n$ obtained in A1,

$$\Delta\varphi = \frac{q_e \left(N_a d_p^2 + \frac{N_a^2}{N_d} d_p^2\right)}{2\varepsilon\varepsilon_0} = \frac{q_e N_a (N_a + N_d) d_p^2}{2\varepsilon\varepsilon_0 N_d}.$$

Similarly,

$$\Delta\varphi = \frac{q_e \left(\frac{N_d^2}{N_a} d_n^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0} = \frac{q_e N_d (N_a + N_d) d_n^2}{2\varepsilon\varepsilon_0 N_a},$$

which gives the answer:

$$\boxed{d_p = \sqrt{\frac{2\varepsilon\varepsilon_0 \Delta\varphi N_d}{q_e N_a (N_a + N_d)}} \approx 58.7\ \text{nm}, \qquad d_n = \sqrt{\frac{2\varepsilon\varepsilon_0 \Delta\varphi N_a}{q_e N_d (N_a + N_d)}} \approx 117.5\ \text{nm}.}$$

A7. From $E_x(x)$ obtained in A3, the absolute value of the electric field is maximal at $x = 0$, where

$$E_{\max} = \frac{q_e N_a d_p}{\varepsilon \varepsilon_0}.$$

Substituting the expression for d_p from A6, we obtain

$$\boxed{E_{\max} = \sqrt{\frac{2 q_e \Delta\varphi N_a N_d}{\varepsilon \varepsilon_0 (N_a + N_d)}} \approx 9.08 \cdot 10^6 \text{ V/m}.}$$

B1. From the distribution of $n(x)$,

$$\frac{n(x = +\infty)}{n(x = -\infty)} = \exp\left(-\frac{U_n(x = +\infty) - U_n(x = -\infty)}{kT}\right)$$

Before the voltage is applied, $n(x = +\infty) = N_d$ and $n(x = -\infty) = n_{10}$. When the voltage is applied, the potential energy difference of an electron $U_n(x = +\infty) - U_n(x = -\infty)$ changes by $(-q_e) \cdot (-V) = q_e V$. According to the problem statement, the electron concentration in the n-type semiconductor is N_d regardless of the voltage. Hence after the voltage is applied still $n(x = +\infty) = N_d$, but $n(x = -\infty) = n_1$. Therefore

$$\frac{N_d}{n_1} = \exp\left(-\frac{q_e V}{kT}\right) \cdot \frac{N_d}{n_{10}},$$

and hence

$$n_1 = n_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

The expression for p_1 is obtained in the same way. From the distribution of $p(x)$,

$$\frac{p(x = +\infty)}{p(x = -\infty)} = \exp\left(-\frac{U_p(x = +\infty) - U_p(x = -\infty)}{kT}\right).$$

Before the voltage is applied, $p(x = +\infty) = p_{10}$ and $p(x = -\infty) = N_a$. When the voltage is applied, the potential energy difference of a hole $U_p(x = +\infty) - U_p(x = -\infty)$ changes by $q_e \cdot (-V) = -q_e V$. According to the problem statement the hole concentration in the p-type semiconductor is N_a, so after the voltage is applied still $p(x = -\infty) = N_a$, but $p(x = +\infty) = p_1$. Therefore

$$\frac{p_1}{N_a} = \exp\left(\frac{q_e V}{kT}\right) \cdot \frac{p_{10}}{N_a},$$

and hence

$$p_1 = p_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

$$\boxed{n_1 = n_{10} \cdot \exp\left(\frac{q_e V}{kT}\right), \qquad p_1 = p_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).}$$

B2. Let A be the cross-sectional area of the n-type semiconductor. Let us calculate the change of the number of holes in the small volume considered during a short time Δt. The change due to recombination is $\left(\frac{dp_1}{dt}\right)_{\text{rec}} A\Delta x\Delta t$. The changes due to diffusion through the two planes are $J_p(x)A\Delta t$ (at the coordinate x) and $-J_p(x+\Delta x)A\Delta t$ (at $x+\Delta x$). The total number of holes, and hence their concentration, is constant, so the total change must be zero:

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} A\Delta x\Delta t + J_p(x)A\Delta t - J_p(x+\Delta x)A\Delta t = 0.$$

Therefore

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = \frac{J_p(x+\Delta x) - J_p(x)}{\Delta x} = \frac{dJ_p(x)}{dx} = -D_p\frac{d^2p_1}{dx^2}.$$

Δp and p_1 differ only by a constant, so

$$\frac{d^2p_1}{dx^2} = \frac{d^2\Delta p}{dx^2}.$$

Taking into account that

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = -\frac{\Delta p}{\tau_p},$$

we obtain

$$\boxed{\frac{d^2\Delta p}{dx^2} = \frac{\Delta p}{\tau_p D_p}}$$

B3. The general solution of the equation obtained in B2 has the form

$$\Delta p = a\exp\left(-\frac{x}{\sqrt{\tau_p D_p}}\right) + b\exp\left(\frac{x}{\sqrt{\tau_p D_p}}\right),$$

where a and b are coefficients. As $x \to +\infty$ the hole concentration must approach the equilibrium value p_{10}, so $\Delta p \to 0$ and $b = 0$. Therefore

$$\Delta p \sim \exp\left(-\frac{x}{\sqrt{\tau_p D_p}}\right),$$

and Δp changes e times when x changes by

$$\boxed{L_p = \sqrt{\tau_p D_p} \approx 2.24\cdot 10^{-5}\ \text{m}.}$$

B4. From the expression for Δp obtained in B3, for $x > d_n$

$$p_1(x) = p_{10} + a\exp\left(-\frac{x}{L_p}\right),$$

where a is some coefficient. Inside the depletion layer the concentration $p(x)$ follows the Boltzmann distribution. Therefore

$$\frac{p(d_n)}{p(-d_p)} = \exp\left(-\frac{U_p(d_n) - U_p(-d_p)}{kT}\right).$$

According to the problem statement the electric field outside the depletion layer can be taken as zero, so the whole voltage drop occurs within $-d_p \le x \le d_n$. Hence, when the voltage is applied, the potential energy difference of a hole $U_p(d_n) - U_p(-d_p)$ changes by $-q_eV$, and as in B1 we can write

$$\frac{p_1(d_n)}{N_a} = \exp\left(\frac{q_eV}{kT}\right)\frac{p_{10}}{N_a},$$

from which

$$p_1(d_n) = \exp\left(\frac{q_eV}{kT}\right)p_{10} = p_{10} + a\exp\left(-\frac{d_n}{L_p}\right),$$

so that

$$a = p_{10}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(\frac{d_n}{L_p}\right).$$

$$\boxed{p_1(x) = p_{10} + p_{10}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(-\frac{x - d_n}{L_p}\right)}$$

B5. To calculate the hole current, consider the n-type region near the boundary of the depletion layer, where the hole flux is determined by diffusion:

$$\begin{aligned} J_p = -D_p\frac{dp_1}{dx} &= \frac{D_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(-\frac{x - d_n}{L_p}\right)\Bigg|_{x=d_n} \\ &= \frac{D_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right). \end{aligned}$$

The current density is $j_p = q_eJ_p$, so

$$\boxed{j_p = \frac{q_eD_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).}$$

B6. As in B5, the electron current has the form

$$j_n = \frac{q_eD_nn_{10}}{L_n}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).$$

Taking into account that

$$p_{10} = \frac{n_i^2}{N_d}, \qquad n_{10} = \frac{n_i^2}{N_a},$$

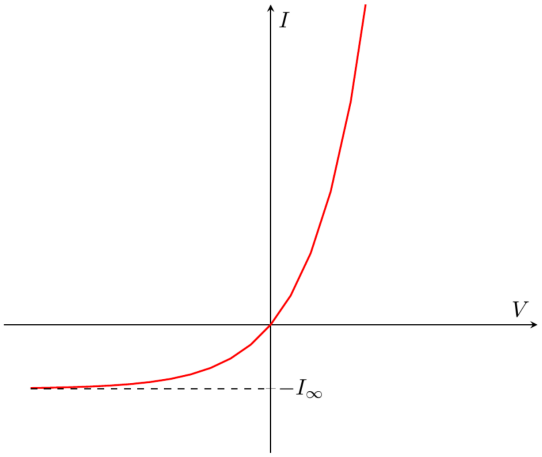
the required current, the sum of the hole and electron currents, is

$$I = q_eAn_i^2\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\left(\frac{D_p}{L_pN_d} + \frac{D_n}{L_nN_a}\right).$$

Using the answer of B3, we obtain

$$\boxed{I = q_e A n_i^2 \left(\exp\left(\frac{q_e V}{kT}\right) - 1\right)\left(\frac{1}{N_d}\sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_a}\sqrt{\frac{D_n}{\tau_n}}\right).}$$

B7.



A qualitative graph of $I(V)$.

B8. The saturation current is

$$I_\infty = q_e A n_i^2 \left(\frac{1}{N_d}\sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_a}\sqrt{\frac{D_n}{\tau_n}}\right) = 5.77 \cdot 10^{-16}\ \text{A},$$

and the current at $V = 0.7$ V is $I_1 = 3.23 \cdot 10^{-4}$ A.

$$\boxed{I_\infty = 5.77 \cdot 10^{-16}\ \text{A}, \qquad I_1 = 3.23 \cdot 10^{-4}\ \text{A}.}$$

> Editor's note: the numbers of the original (marking scheme included) correspond to $n_i = 1.5 \cdot 10^{16}\ \text{m}^{-3}$. With the $n_i = 1.0 \cdot 10^{16}\ \text{m}^{-3}$ given in the problem, $I_\infty \approx 2.56 \cdot 10^{-16}$ A and $I_1 \approx 1.44 \cdot 10^{-4}$ A.